# Paul MacLean

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## Education

**MS in Computer Science**, West Virginia University, Morgantown, WV

**BS in Computer Science,** California University of Pennsylvania, California, PA

## Certifications and Trusts

- AWS Solutions Architect Associate (WIP)
- Public Trust
- SAFe 6.0 Certified Practitioner

## Experience

**Agile 5 Technologies** November 2022 – November 2025

Full Stack Software Engineer

- Extended Ember JS and Spring Boot microservices into new comment clustering features, halving review time and saving thousands of manhours
- Maintained and deployed legacy and modernized systems in Kubernetes in AWS
- Developed a modernized frontend for regulations.gov using SvelteKit and TailwindCSS to serve 220 agencies
- Translated UX wireframes, legacy behavior, and product owner requirements into workable stories, NestJS endpoints, and SvelteKit components
- Ensured product was authorized, tested, branded, 508 compliant, and to-spec
- Designed a rule engine that assesses NIST 800-53 cloud compliance across 20+ AWS services via SDK
- Implemented aforementioned rule engine as a monolithic codebase using Docker, NextJS, and Pocketbase

**West Virginia University** August 2020 – May 2022

Graduate Research Assistant

- Created and implemented an ML approach for predicting programmer effort using Jira and GitHub data
- Engineered Python pipeline to mine, store, structure, and classify data from open-source projects
- Documented design process via technical drawings and reports for NASA IV & V
- Graded Analysis of Algorithms fundamentals (data structures, complexity, sorting, graph algorithms)

**U.S. Steel** May 2019 – September 2019

Process Control Engineer (Internship)

- Learned the inner workings of the steelmaking process, and the integration of technology throughout
- Fixed bugs in a VMS operating system, Fortran, and SQL stack

## Publications

Paul MacLean, Tom Devine, Dale Dzielski, Joshua Hernandez (2023). Efficacy of Reported Issue Times as a Means for Effort Estimation. CSCE 2023, IEEE. ISBN: 979-8-3503-2759-5

## Skills

- **Tools and Concepts:** NodeJS, NestJS, SvelteKit, Tailwind, Spring, AWS, Git, Testing, Playwright, Kubernetes, CI/CD, EmberJS, Docker, Agile/Scrum, Machine Learning
- **Languages:** TypeScript, Python, Java, C, C++, Lua
- **Consulting & Collaboration:** System Thinking, Solution Design, Client Engagement, Troubleshooting, Knowledge Transfer, Jira, Confluence

## Projects

**Othello Boardgame Digitizer**

Trained neural network to classify Othello tiles as empty, white, or black

Applied image processing techniques to transform off-angle board images into 2D arrays for virtual play

**Vulcan Transit Tracker**

Built an IoT prototype for an Arduino bus tracking system for alma mater shuttle service

Interfaced with 4G module via AT commands to send coordinates to backend